

Lakshmi Sarvani Kandala

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EDUCATION

San Jose State University

Bachelor of Science in Computer Engineering

San Jose, CA

May 2024 – Present

EXPERIENCE

AI Intern

Jun 2025 – Aug 2025

Cogent IBS

Remote

- Migrated customer, product, and order data from MongoDB to PostgreSQL using Python/SQL; designed normalized schemas and indexing for transactional performance
- Built and tested ETL pipelines with automated validation (referential integrity, type constraints, deduplication); implemented backend-facing endpoints for real-time lookups
- Benchmarked migration/query performance and improved reliability through iterative transformations; collaborated via Git code reviews and agile development workflows

RESEARCH

Multimodal Real Estate Price Estimation

Jun 2023 – Aug 2023

Blast AI Bootcamp

Remote

- Implemented CNN-based feature extraction in Python and fused image + structured features for predictive modeling
- Built reproducible preprocessing/evaluation pipelines; tracked metrics across experiments to compare model variants
- Applied dimensionality reduction and feature selection to improve stability; documented results with emphasis on repeatability

PROJECTS

Edge AI Smart Entry Assistant | *Embedded Systems, BLE, Python, C++*

Jan 2026 – Present

- Designing a context-aware cyber-physical system that detects multi-user presence near a door and predicts departure intent to trigger personalized reminders
- Architecting the full system (block diagrams, data flow, state machines) and implementing event-driven firmware logic for reliable real-time state transitions
- Integrating hardware sensors (BLE RSSI, proximity/motion, door events, item presence) and implementing on-device signal processing (RSSI smoothing, noise reduction, feature extraction)
- Building a streaming simulation + data logging pipeline to generate/label time-series traces, test edge cases (overlapping users, RSSI noise), and validate timing constraints
- Deploying and optimizing an intent-classification model on ESP32/Raspberry Pi, focusing on inference latency, memory footprint, concurrency, and robustness under wireless variability

TECHNICAL SKILLS

Programming Languages: C++, Python, x86 Assembly, SQL, Verilog HDL

Software Engineering: Data Structures, Algorithms, Software Development, Compiler Design,

Databases: PostgreSQL, MongoDB (NoSQL → SQL Migration), Relational Data Modeling, Normalization, Query Optimization

Embedded / Hardware: Embedded System Architecture, Event-Driven State Machines, Sensor Integration (BLE RSSI, Proximity/Motion), Signal Processing, FPGA Design, RTL Design, Digital Design, CMOS Logic Circuits, Microprocessor Architecture, Circuit Analysis, Operational Amplifiers, ADC/DAC Converters

Networking: Computer Networks, TCP/IP Layers, Routing, Network Protocols, Data Link and Physical Layer Communication

Tools: Git, Jupyter Notebook, Google Colab, Visual Studio Code, MATLAB, FPGA CAD Tools

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Architecture, Digital Design, Circuit Analysis, Engineering Probability and Statistics, x86 Processor Assembly Language, Compiler Design, Electronics for Computing Systems, Microprocessor Design, Computer Networks